WORKED SOLUTION

STEP II 1993, Question 2

Pure • Trigonometric and Algebraic Integrals

> **STEP II 1993, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/93-s2-q2

The key idea

In part (i) a product of cosines is awkward to integrate, but a sum is easy, so we turn the product into a sum with the factor formula. Everything then depends on one fact: $\cos kx$ integrates to zero over a full period unless $k = 0$. The special cases are exactly the ones where $m - n$ or $m + n$ is zero. In part (ii) we combine the square root into a single fraction and rationalise, which splits the integrand into two standard pieces.

Solution

Part (i)

The factor formula $2\cos A\cos B = \cos(A - B) + \cos(A + B)$ gives

$$\cos(mx)\cos(nx) = \tfrac{1}{2}[\cos((m-n)x) + \cos((m+n)x)].$$

So we need $\int_0^{2\pi} \cos(kx)\,\mathrm{d}x$ for an integer k:

$$\int_0^{2\pi} \cos(kx)\,\mathrm{d}x = \begin{cases} \left[\dfrac{\sin kx}{k}\right]_0^{2\pi} = \dfrac{\sin 2k\pi}{k} = 0, & k \neq 0, \\ \displaystyle\int_0^{2\pi} 1\,\mathrm{d}x = 2\pi, & k = 0. \end{cases}$$

Here we used that k is an integer, so $\sin 2k\pi = 0$. We can't divide by k when $k = 0$, which is why that case is separate.

> **Ask yourself** How many different cases are there, and do they overlap?

Each of $k = m - n$ and $k = m + n$ is either zero or not, and both are zero only when $m = n = 0$. That gives three cases:

- $m \neq n$ and $m \neq -n$, that is $|m| \neq |n|$: both integrals vanish and the answer is 0.

- Exactly one of $m-n$, $m+n$ is zero, that is $|m| = |n| \neq 0$: one term contributes $\frac{1}{2} \cdot 2\pi$ and the other 0, so the answer is π.
- $m = n = 0$: both terms contribute $\frac{1}{2} \cdot 2\pi$. (Directly, the integrand is 1.) The answer is 2π.

In summary,

$$\int_0^{2\pi} \cos(mx)\cos(nx)\,\mathrm{d}x = \begin{cases} 0, & |m| \neq |n|, \\ \pi, & |m| = |n| \neq 0, \\ 2\pi, & m = n = 0. \end{cases}$$

Note that $m = -n$ belongs with $m = n$, since cos is even and $\cos(-nx) = \cos(nx)$.

Part (ii)

The integrand $\sqrt{1+1/x} = \sqrt{(x+1)/x}$ is real only when $(x+1)/x \geq 0$, that is for $x > 0$ or $x \leq -1$. We deal with $x > 0$ first.

The case $x > 0$. Multiply top and bottom inside the root by $x+1$:

$$\sqrt{\frac{x+1}{x}} = \sqrt{\frac{(x+1)^2}{x(x+1)}} = \frac{x+1}{\sqrt{x^2+x}},$$

using $x+1 > 0$ to take the square root of $(x+1)^2$. Now split the numerator so that one part is half the derivative of x^2+x:

$$\frac{x+1}{\sqrt{x^2+x}} = \frac{1}{2} \cdot \frac{2x+1}{\sqrt{x^2+x}} + \frac{1}{2} \cdot \frac{1}{\sqrt{x^2+x}}.$$

? **Ask yourself** How do we integrate $1/\sqrt{x^2+x}$?

We complete the square: $x^2 + x = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4}$. Then the standard result $\int (u^2 - a^2)^{-1/2}\,\mathrm{d}u = \ln\left(u + \sqrt{u^2 - a^2}\right)$, valid for $u > a > 0$, applies with $u = x + \frac{1}{2}$ and $a = \frac{1}{2}$:

$$\int \frac{\mathrm{d}x}{\sqrt{x^2+x}} = \ln\left(x + \tfrac{1}{2} + \sqrt{x^2+x}\right) + \text{const} = \ln\left(2x + 1 + 2\sqrt{x^2+x}\right) + \text{const}.$$

The first piece is $\int \frac{1}{2}(2x+1)(x^2+x)^{-1/2}\,\mathrm{d}x = \sqrt{x^2+x}$. So, for $x > 0$,

$$\int \sqrt{1 + \frac{1}{x}}\,\mathrm{d}x = \sqrt{x^2+x} + \frac{1}{2}\ln\left(2x + 1 + 2\sqrt{x^2+x}\right) + C.$$

Since $2x + 1 + 2\sqrt{x^2+x} = \left(\sqrt{x} + \sqrt{x+1}\right)^2$, this can also be written as

$$\sqrt{x(x+1)} + \ln\left(\sqrt{x} + \sqrt{x+1}\right) + C.$$

The substitution $x = \sinh^2 t$ gives this second form directly. The integrand becomes $2\cosh^2 t = 1 + \cosh 2t$, which integrates to $t + \sinh t \cosh t$.

The case $x \le -1$. Now $x+1 \le 0$, so $\sqrt{(x+1)^2} = -(x+1)$ and

$$\sqrt{\frac{x+1}{x}} = -\frac{x+1}{\sqrt{x^2+x}}.$$

The same split works, but now $x + \frac{1}{2} < 0$, so we need $\int (u^2 - a^2)^{-1/2}\,\mathrm{d}u = \ln\left|u + \sqrt{u^2-a^2}\right|$, valid for $|u| > a$. For $x < -1$,

$$\int \sqrt{1 + \frac{1}{x}}\,\mathrm{d}x = -\sqrt{x^2+x} - \frac{1}{2}\ln\left|2x+1+2\sqrt{x^2+x}\right| + C.$$

Because $\left(2x+1+2\sqrt{x^2+x}\right)\left(2x+1-2\sqrt{x^2+x}\right) = 1$, the logarithm can also be written as $+\frac{1}{2}\ln\left|2x+1-2\sqrt{x^2+x}\right|$.

Checks

- Part (i): with $m = n = 1$ the integrand is $\cos^2 x$, whose average is $\frac{1}{2}$, so the integral is $\frac{1}{2} \cdot 2\pi = \pi$, as predicted. A numerical (Simpson's rule) evaluation for every pair $-6 \le m, n \le 6$ matched the three-case formula.
- Part (ii): differentiate the answer. The derivative of $\sqrt{x^2+x}$ is $\dfrac{2x+1}{2\sqrt{x^2+x}}$. The derivative of $\frac{1}{2}\ln(\cdots)$ is $\dfrac{1}{2\sqrt{x^2+x}}$. They add to $\dfrac{x+1}{\sqrt{x^2+x}}$, as required.
- Numerically, $\int_1^3 \sqrt{1+1/x}\,\mathrm{d}x \approx 2.48547$, both by Simpson's rule and from the formula: $\sqrt{12} - \sqrt{2} + \frac{1}{2}\ln\dfrac{7+4\sqrt{3}}{3+2\sqrt{2}}$.

Common slips

- Forgetting the case $m = -n \neq 0$, which gives π just as $m = n \neq 0$ does.
- Writing $\dfrac{\sin kx}{k}$ when $k = 0$; that case needs separate treatment.
- Writing $\sqrt{(x+1)^2} = x+1$ without checking the sign, which gives the wrong answer for $x < -1$.

Beyond STEP

Part (i) says that the functions $1, \cos x, \cos 2x, \ldots$ are *orthogonal* on $[0, 2\pi]$. This is the fact behind Fourier series: if $f(x) = \sum a_n \cos nx$, multiplying by $\cos mx$ and integrating picks out the single coefficient a_m. The same idea, with an integral playing the part of a dot product, underlies much of linear algebra on spaces of functions. Strang's *Introduction to Linear Algebra* covers orthogonality, and Boyce and DiPrima's *Elementary Differential Equations* shows Fourier series in action solving the heat equation.

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